Indications

Amiodarone is one of the most commonly used anti-arrhythmic drugs. While the United States FDA has labeled amiodarone for the treatment of life-threatening ventricular arrhythmias, the drug is commonly used off-label to treat supraventricular tachyarrhythmias such as atrial fibrillation as well as for the prevention of ventricular tachyarrhythmias (VTs) in high-risk patients.[1][2][3]

One of the most common indications for amiodarone use in the acute setting is atrial fibrillation with a rapid ventricular response. It is particularly useful in hemodynamically unstable patients and patients with congestive heart failure with reduced left ventricular ejection fraction who may be adversely affected by negative inotropic or vasodilating effects of other rate-controlling agents. Amiodarone has effectively controlled the ventricular rate and converted to and maintained sinus rhythm in patients with atrial fibrillation and rapid ventricular response.[2]

Amiodarone also can be used to treat other supraventricular tachyarrhythmias, including atrial flutter, refractory AV (atrioventricular) nodal, and AV re-entrant tachycardia (commonly referred to as SVT). Finally, amiodarone is indicated for the treatment of ventricular arrhythmias, specifically monomorphic VT, non-Torsades polymorphic VT (secondary to myocardial ischemia and not associated with prolonged QTc), as well as for pulseless ventricular fibrillation (VF) and pulseless VT that fail to convert after CPR, defibrillation, and epinephrine administration. When studied in patients with out-of-hospital cardiac arrest, amiodarone resulted in a higher rate of return of spontaneous circulation (ROSC); however, this ROSC benefit did not result in a higher rate of survival to hospital discharge with the favorable neurological outcome.[3]

Mechanism of Action

Amiodarone is primarily a class III antiarrhythmic. Like other antiarrhythmic drugs of this class, amiodarone works primarily by blocking potassium rectifier currents responsible for the repolarization of the heart during phase 3 of the cardiac action potential. This potassium channel-blocking effect results in increased action potential duration and a prolonged effective refractory period in cardiac myocytes. Myocyte excitability is decreased, preventing reentry mechanisms and ectopic foci from perpetuating tachyarrhythmias. Electrocardiographic evidence of these effects is evident as prolongation of the QRS duration and QTc interval.[4]

Unlike other class III agents, amiodarone also blocks beta-adrenergic receptors (e.g., beta-1), calcium channels, and sodium channels. Further electrophysiological manifestations of this drug effect include decreased SA (sinoatrial) node automaticity and AV node conduction velocity and inhibited ectopic pacemaker automaticity. In some instances, these additional mechanisms of action can result in unwanted side effects, including bradycardia, hypotension, and Torsades de pointes (TdP).[5]

Amiodarone has a very long half-life (t1/2 of several weeks), and it may take 6 weeks for full clinical effects with oral therapy. Also, upon discontinuation of amiodarone therapy, pharmacologic effects could continue for 1 to 3 months.[6]

Administration

In adult patients, amiodarone can be given for VT/VF cardiac arrest via intravenous (IV)/intraosseous (IO) infusion as a 300 mg rapid bolus followed by an additional bolus of 150 mg IV/IO if VT or VF persists. If the patient achieves ROSC, a continuous infusion is started at 1 mg/min for 6 hours, followed by 0.5 mg/min. To treat all acute tachyarrhythmias in adults, amiodarone can be given IV 150 mg over 10 minutes, followed by a 1 mg/min infusion for 6 hours, followed by an infusion at 0.5 mg/min. The recommended total dose over 24 hours should not exceed 2.4 grams. Recommended oral dosing is 400 to 600 mg daily in divided doses for 2 to 4 weeks, followed by maintenance dosing of 100 to 200 mg daily. No dosage adjustment is necessary for renal impairment.

Pediatric advanced life support dosing is 5 mg/kg (maximum 300 mg per dose) rapid bolus IV/IO for cardiac arrest. This dose may be repeated two times if VF or pulseless VT persists. For perfusing pediatric tachyarrhythmias, the patient receives a similar loading dose; however, the time of administration stretches to 20 to 60 minutes - this can be followed by a 5 mcg/kg/min infusion, which may be titrated up to a max dose of 15 mcg/kg/min for a max of no more than 20 mg/kg within 24 hours.

Adverse Effects

The prevalence of adverse effects from amiodarone therapy is as high as 15% within the first year of use and 50% for long-term use. The risk/benefit ratio often leads to the discontinuation of amiodarone in the first year of treatment for patients with atrial fibrillation. The most common side effect is corneal microdeposits, which occur in at least 90% of patients taking amiodarone; this is thought to be due to the secretion of amiodarone in the lacrimal gland with uptake by the corneal epithelium. However, only about 10% of these patients will develop actual visual symptoms. Other ocular side effects include photophobia, optic neuropathy, and visual halos. Baseline ophthalmologic evaluation is recommended for patients starting amiodarone.[7][8]

Cardiac toxicity from amiodarone administration can be related to its therapeutic mechanism. Unfortunately, the same qualities that make this drug useful for treating arrhythmias can also lead to bradycardia and atrioventricular and intraventricular conduction abnormalities. While most studies have proven the safety of amiodarone regarding pro-arrhythmic events, amiodarone may induce TdP in the first 48 hours of IV administration. Amiodarone may be the most common cause of drug-induced TdP. The risk of developing TdP increases in patients who have pre-existing electrolyte abnormalities or are also receiving treatment with beta-blockers and/or digoxin. QTc prolongation is a risk factor for TdP, but amiodarone-induced TdP can occur even with a normal QTc interval. Chronic oral administration of amiodarone is rarely associated with TdP.[9]

Pulmonary toxicity typically presents in the first year of use and most commonly resembles interstitial lung disease. However, pulmonary toxicity also can present as organizing pneumonia, pleural effusion, acute respiratory distress syndrome, or diffuse alveolar hemorrhage. Unfortunately, there is no pathognomonic finding to diagnose amiodarone-induced pulmonary toxicity, and work up involves rigorous evaluation to exclude alternative diagnoses. Mortality from amiodarone-induced pulmonary toxicity has been reported to be close to 10%. Fortunately, steroids have been useful in treating this adverse reaction. Baseline and yearly chest X-rays are recommended for patients starting amiodarone, and in some cases, pulmonary function tests may be indicated to assess for the development of pulmonary toxicity.[10]

Amiodarone therapy may result in hypo- or hyperthyroidism, with hypothyroidism being almost twice as common. Toxicity usually is related to thyroiditis. Due to the risk of thyroid disease, the clinician should obtain baseline thyroid function tests and followed up every six months. Amiodarone-induced thyroid toxicity should be suspected in patients with weight loss or any change in cardiac status. Treatment of hypothyroidism includes levothyroxine. Treatment of hyperthyroidism may involve a combination of corticosteroids, propylthiouracil or methimazole, and possible thyroidectomy. Patients with a pre-existing hyperthyroid disease can develop thyrotoxicosis because amiodarone contains iodine. Amiodarone-associated thyrotoxicosis can be difficult to treat and carries a high risk of mortality. Adding to this complexity is amiodarone's beta-blocking effects, which can mask classic symptoms of thyrotoxicosis.[11]

There is a 1% annual incidence of liver toxicity in patients treated with amiodarone. Most cases resolve after stopping the drug; however, toxicity can occasionally progress to end-stage liver disease and cirrhosis. IV amiodarone may cause acute liver injury within one day of infusion. Nausea, anorexia, and constipation are the most common gastrointestinal side effects.

Neurologic toxicity can occur in up to 27.5% of patients, ranging from cognitive impairment to peripheral neuropathy, ataxia, and in some rare cases, quadriplegia. Dermatologic effects include blue skin discoloration ("smurf-skin") and photosensitivity. In rare instances, amiodarone may cause epididymitis and erectile dysfunction.

Drug interactions are another adverse effect of amiodarone. Given that amiodarone is a cytochrome P450 inhibitor, it can reduce warfarin clearance. The international normalized ratio (INR) should be monitored closely in patients starting amiodarone or have a change in the dosing of either amiodarone or warfarin. Digoxin concentrations have increased, even doubled, with co-administration of amiodarone. Prescribing any QTc-prolonging agent in addition to amiodarone should be carefully considered, and the QTc interval should be checked regularly after starting the second agent.[12]

Peripheral IV administration of amiodarone can result in phlebitis. If a patient has a central venous catheter, that route may be preferable to peripheral IV. When infusing amiodarone through a peripheral IV, the access site requires frequent monitoring, and the infusion stopped and changed to another site if the staff notes phlebitis.

Contraindications

Amiodarone is contraindicated in patients with second or third-degree heart block who do not have pacemakers. Amiodarone contraindications also include patients with pre-excitation (Wolff-Parkinson-White syndrome) and concurrent atrial fibrillation. Amiodarone should be avoided in patients with baseline QTc prolongation. The incidence of hypersensitivity reactions to amiodarone in patients who have documented iodine allergies should be carefully evaluated; however, retrospective studies have concluded that allergies to iodine or iodinated contrast agents may not constitute an absolute contraindication to amiodarone.[13]

Hyperkalemia and sodium channel blocker toxicity, such as in tricyclic antidepressant, bupivacaine, and quinidine toxicity, may produce arrhythmias that resemble VT. Accelerated idioventricular rhythm also may resemble VT. Treating with amiodarone in these clinical situations can precipitate a hemodynamic collapse. While these clinical situations are not commonly listed under contraindications, it is crucial to consider these arrhythmias within the correct clinical context to avoid cardiovascular deterioration.[14]

Administering amiodarone to treat atrial fibrillation with rapid ventricular response involves complex decision-making and broad differential diagnosis. Due to amiodarone's adverse side effect profile, alternative therapies should merit consideration before administering amiodarone. Also, the clinician should ensure that the patient has no reversible causes, such as fever, hypovolemia, or hypoxia, for atrial fibrillation with a rapid ventricular response before initiating treatment with amiodarone. For these patients, addressing the underlying cause should be the prime mode of therapy rather than administering amiodarone.

Monitoring

In the acute care setting, patients receiving amiodarone will need continuous cardiac monitoring and should receive frequent reassessment of vital signs and clinical conditions.[15][16]

Per the American Heart Association and Advanced Cardiac Life Support guidelines, patients who are hemodynamically unstable from tachyarrhythmias more rapid than 150 beats per minute should receive direct current cardioversion. If cardioversion is unsuccessful, these patients may receive treatment with amiodarone. Hemodynamically unstable patients with heart rates less than 150 beats per minute need to have other etiologies explored as the cause of their instability. However, treatment with cardioversion or amiodarone may still be necessary.

Anticoagulation is a consideration in all patients receiving amiodarone for atrial fibrillation who have no clear contraindications. A CHA2DS2-VASc score may aid clinical decision-making regarding the initiation of anticoagulation.

Toxicity

Acute amiodarone toxicity from overdose is a relatively rare phenomenon. Its high volume of distribution makes it unlikely to cause serious toxicity as single ingestion. However, patients with suspected acute amiodarone overdose should have continuous cardiac monitoring for two to three days after ingestion as toxic effects may experience a delay before appearing. The literature on toxicity from acute amiodarone overdose describes primarily cardiovascular side effects, including hypotension, bradycardia, VT, and TdP. However, as noted above, these adverse effects may occur at recommended dosing as well.[17]

Treatment of acute toxicity may include vasopressor support as well as magnesium for TdP. Temporary pacing may be necessary for bradycardia, and TdP treatment may warrant over-drive pacing. Activated charcoal is the recommended therapy for acute oral ingestions. No specific antidote for amiodarone toxicity is available, and amiodarone is not dialyzable.

Enhancing Healthcare Team Outcomes

Any clinician (MD, DO, PA, NP) should consult with a cardiologist before starting a patient on amiodarone. The drug has many serious side effects, and there is also the potential for interacting adversely with other medications. Patients receiving amiodarone should chronically have baseline testing done to facilitate future monitoring of hepatic, thyroid, pulmonary, and ophthalmologic toxicity. Patients on warfarin will also need close monitoring of their INR. The provider and pharmacist should perform thorough medication reconciliation and consult point-of-care medical resources for potential drug interactions that may require dosage adjustment.

Drug interactions with amiodarone may include but are not limited to digoxin, procainamide, diltiazem, verapamil, beta-blockers, phenytoin, warfarin, and statins. It is also wise to include a cardiology board-certified pharmacist when starting amiodarone therapy to verify therapeutic appropriateness, dosing, and drug interactions. Nursing staff with cardiology specialty training should also be aware of amiodarone side effects and assist in evaluating patient adherence and therapeutic effectiveness. All providers, including clinicians, specialists, nursing, and pharmacy, must operate as an interprofessional team when considering amiodarone therapy so that the drug can provide maximum benefit with minimal adverse events. [Level 5]